

Final Exam CSC 240 (16-Dec-2025)

1. What is an abstraction in the context of programming, and give an example from C++?

An abstraction simplifies complex systems by hiding details; e.g., the `int` data type abstracts integer numbers.

2. Name the three main components of the Standard Template Library (STL) in C++.

Containers, iterators, and algorithms.

3. What is the difference between sequence containers and associative containers in STL?

Sequence containers store data sequentially (e.g., `vector`); associative containers store by key for fast lookup (e.g., `map`).

4. Explain Big-O notation briefly and give an example for linear time complexity.

Big-O describes the upper bound of an algorithm's time complexity; $O(n)$ means time grows linearly with input size, like traversing an array.

5. How do you declare a pointer in C++, and what does the dereference operator do?

Declare with `int* ptr;`; the `*` operator accesses the value at the pointer's address.

6. What is the main difference between procedural and object-oriented programming?

Procedural focuses on functions and procedures; OOP focuses on objects, classes, and message passing between objects.

7. Describe a default constructor for a vector in STL and what it creates.

`vector<string> names;` creates an empty vector of strings.

8. What is dynamic memory allocation in C++, and how do you do it with pointers?

Allocating memory at runtime using `new`; e.g., `int* ptr = new int;`

9. Why are iterators important in STL?

They provide uniform access to container elements, acting like pointers without needing indices.

10. Give an example of quadratic time complexity $O(n^2)$ and explain why it occurs.

Nested loops, like comparing every pair in an array; time grows with the square of input size due to double iteration.